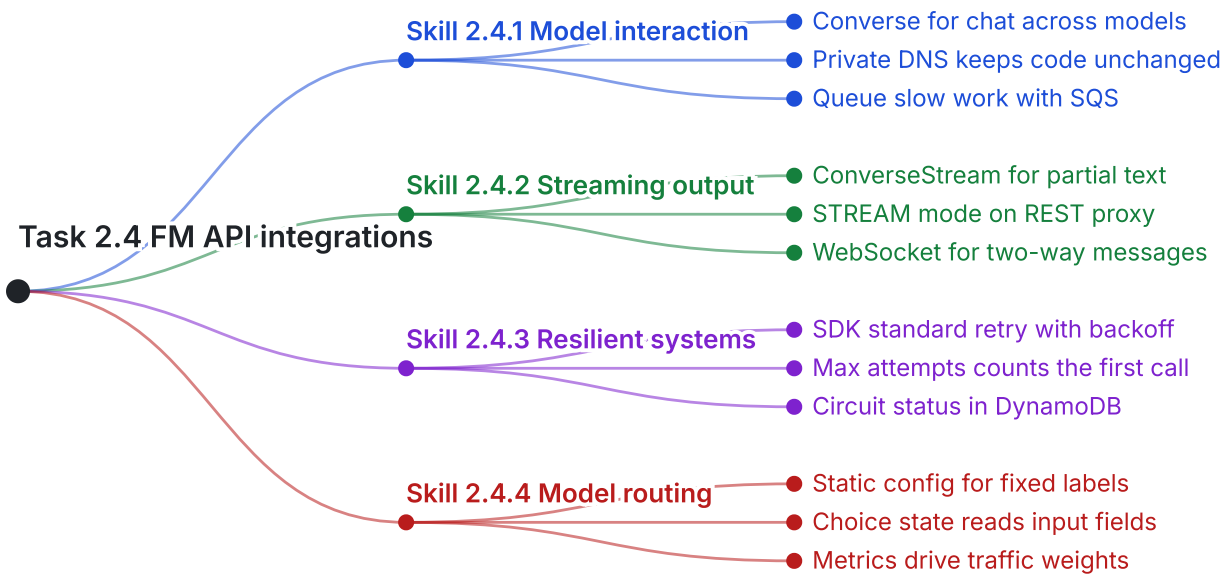
Memory sheet: Task 2.4 Implement FM API integrations

Domain 2, weight 26 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
Streamed text must reach the user through API Gateway	REST API with STREAM mode and a proxy integration	The API is an HTTP API, which has no response streaming
The browser and the service both send messages in one session	WebSocket API with a route for each message type	Text flows one way only, where a REST stream or server-sent events is simpler
One chat app must use one message format across several models	Converse or ConverseStream on bedrock-runtime	The request body must match one model family, which points to InvokeModel
The caller can take a job ID now and the answer later	SQS queue with a worker and a dead-letter queue	The answer must return in the same HTTP response
A private subnet must reach Bedrock with no public internet path	Interface endpoint for bedrock-runtime with private DNS on	A NAT gateway path, which still crosses the internet gateway
Brief throttles or transient errors hit individual calls	SDK retry mode set to standard, with a capped max attempts setting	The dependency stays down for minutes, where more retries add load
One dependency keeps failing across many callers or executions	Circuit state stored outside the call, with a fallback branch	Brief errors that a retry clears on the next attempt
The model branch depends on a field in each request	Step Functions Choice state with a Default branch	A tenant label that never changes, where a plain config map is simpler

Etch lines

1. Streamed replies need a REST API with a proxy integration and STREAM mode.
2. Keep the circuit record shared across executions, so each one sees the failure.
3. Send slow work to SQS when the caller accepts a later result.
4. An interface endpoint with private DNS on keeps the SDK code unchanged.
5. A NAT gateway path still uses the internet gateway, which the rule forbids.
6. Max attempts counts the first call, so a value of 3 allows two retries.

Flash questions

- Q: Which API type can stream a model reply through a proxy integration? A: A REST API with STREAM transfer mode, since HTTP APIs lack response streaming.
- Q: Which front door fits a browser that must send messages while the server also pushes updates? A: A WebSocket API with a route for each message type.
- Q: What does the max attempts setting count? A: The first call plus retries, so a value of 3 allows two retries.
- Q: Which network option keeps bedrock-runtime traffic off every internet path? A: An interface endpoint with private DNS on, not a NAT gateway.
- Q: A caller can take a job ID now and the answer later. Which pattern fits? A: Accept the request, queue it in SQS, and let a worker call the model.
- Q: Does a 429 from API Gateway prove that Bedrock quota ran out? A: No, the gateway returned it before the backend ran.
- Q: Where should circuit state live when many executions call one failing dependency? A: In shared storage such as DynamoDB, which a Choice state reads before each call.
- Q: Which Step Functions state tests request fields to pick a model branch? A: A Choice state, which needs a Default branch for unmatched input.

Source: Task 2.4 sections for Skill 2.4.1 to 2.4.4, the X versus Y pairs, Read a long question, and Drill. AWS docs and the study guide sections above, no practice-test text.